

EduPath AI - Major Recommendation System

Ethical Impact Assessment

1. Introduction & Project Objectives

Selecting a university major is a crucial decision that significantly determines a student's academic path and future career. The vast array of major choices, limited self-awareness, and external social expectations often lead to inappropriate major selections that can be detrimental.

EduPath AI is proposed as an objective decision support tool utilizing Artificial Intelligence. This Ethical Impact Assessment (EIA) report is compiled to ensure that the design, development, and utilization of this application consistently adhere to artificial intelligence ethical standards, prioritizing user safety, safeguarding data privacy, and maintaining complete human control.

2. CORE ETHICAL ANALYSIS

2.1 Privacy and Data Minimization

Ethical Statement: Student data is processed only with user consent and is not permanently stored without authorization.

Technical Evidence: The REST API endpoint `/api/analyze` in `app.py` processes the incoming student profile request in temporary local memory variables (`nilai_dict`, `riasec_dict`, `prestasi`). There are no database import statements (e.g., SQL/NoSQL libraries) or write commands anywhere in the codebase. All session data is completely purged upon browser refresh.

2.2 Fairness and Non-Discrimination

Ethical Statement: The system provides recommendations objectively without discrimination based on gender, socioeconomic background, or other personal characteristics.

Technical Evidence: In `model/knn_model.py`, the input feature vector for similarity calculation is strictly limited to the `FEATURE_COLS` list:

```
FEATURE_COLS = [
    'math', 'indonesian', 'english', 'biology', 'physics', 'chemistry',
    'economics', 'sociology',
    'R', 'I', 'A', 'S', 'E', 'C'
```

Sensitive demographic features (such as gender, race, or family income) are completely omitted from the model inputs, ensuring demographic parity.

2.3 Transparency and Explainability (Explainable AI)

Ethical Statement: Every recommendation is accompanied by a clear explanation of the factors influencing the AI's decision.

Technical Evidence: The server does not return arbitrary outputs. It runs `generate_personal_alasan` from `backend/utlis.py`, which dynamically matches the student's highest academic grades and dominant RIASEC traits with the recommended major's

requirements. This reasoning is returned in the API response JSON object under the **alasan** key and displayed directly on the UI.

2.4 Human Oversight and Agency

Ethical Statement: Students are encouraged to consult with school counselors before making final educational decisions.

Technical Evidence: The user interface in **frontend/index.html** implements explicit call-out banners and alert messages advising students that the results are supportive recommendations, prompting them with a direct instruction to consult with their school counselor (Guru BK) before making enrollment decisions.

3. System Ethical Architecture Detail

The ethical implementations mentioned above are strictly enforced within the system's processing architecture workflow as follows:

- **INPUT:** Accepts academic scores (grades in 8 subjects), interests & extracurricular achievements, and answers from the 36-question RIASEC assessment.
- **PRE-PROCESSING:** Performs feature extraction, calculates RIASEC personality scores (Realistic, Investigative, Artistic, Social, Enterprising, Conventional), and normalizes features into a uniform scale of 0 to 1. This normalization prevents high-range features (such as 0–100 academic scores) from dominating calculations over lower-range features (such as 6–30 RIASEC scores).
- **AI RECOMMENDATION ENGINE:** Processes student data using the K-Nearest Neighbors (KNN) algorithm with $K=5$ (or $K=7$) assisted by the KD-Tree search algorithm to mathematically match student profiles with the 5 closest major metric profiles, followed by major ranking.
- **OUTPUT:** Instantly presents the Top 3 Major Recommendations along with match similarity percentages and rational explanations on the user interface screen.